

Welcome to

Become a Python Developer

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A simple approach is to do a **linear search**, i.e

1. Start from the leftmost element of arr[] and one by one compare x with each element of arr[]
2. If x matches with an element, return the index.
3. If x doesn't match with any of the elements, return -1.

#linear search

```
def search(arr, x):
```

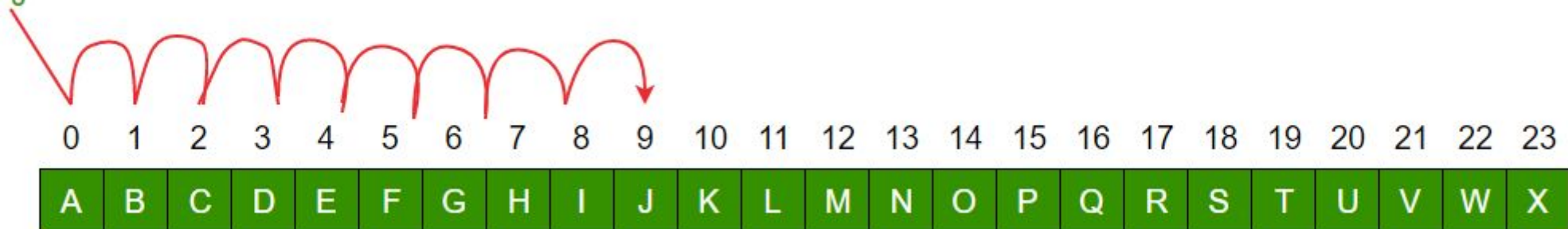
```
    for i in range(len(arr)):
```

```
        if arr[i] == x:
            return i
```

```
    return -1
```

Visualization

Find 'J'



Binary Search

a searching algorithm for finding an element's position in a sorted array

Algorithm:

do until the pointers low and high meet each other.

mid = (low + high)/2

if (x == arr[mid])

return mid

else if (x > arr[mid]) // x is on the right side

low = mid + 1

else // x is on the left side

high = mid - 1

Binary Search in python

def binarySearch(array, x, low, high):

Repeat until the pointers low and high meet each other

while low <= high:

mid = low + (high - low)//2

if array[mid] == x:

return mid

elif array[mid] < x:

low = mid + 1

else:

high = mid - 1

return -1

array = [3, 4, 5, 6, 7, 8, 9]

x = 4

result = binarySearch(array, x, 0, len(array)-1)

if result != -1:

print("Element is present at index " + str(result))

else:

print("Not found")

Bubble Sort

Bubble sort is a sorting algorithm that compares two adjacent elements and swaps them until they are in the intended order.

step = 0

i = 0



i = 1



i = 2



i = 3



step = 1

i = 0



i = 1



i = 2



Bubble Sort

step = 2

i = 0



i = 1



step = 3

i = 0



Selection Sort

Selection sort is a sorting algorithm that selects the smallest element from an unsorted list in each iteration and places that element at the beginning of the unsorted list.

Step 1 – Set MIN to location 0

Step 2 – Search the minimum element in the list

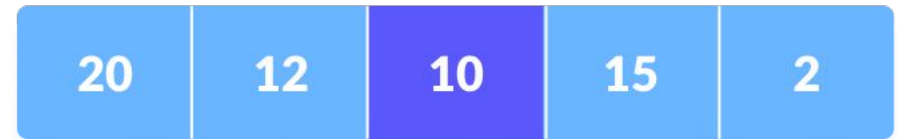
Step 3 – Swap with value at location MIN

Step 4 – Increment MIN to point to next element

Step 5 – Repeat until list is sorted



First element as minimum



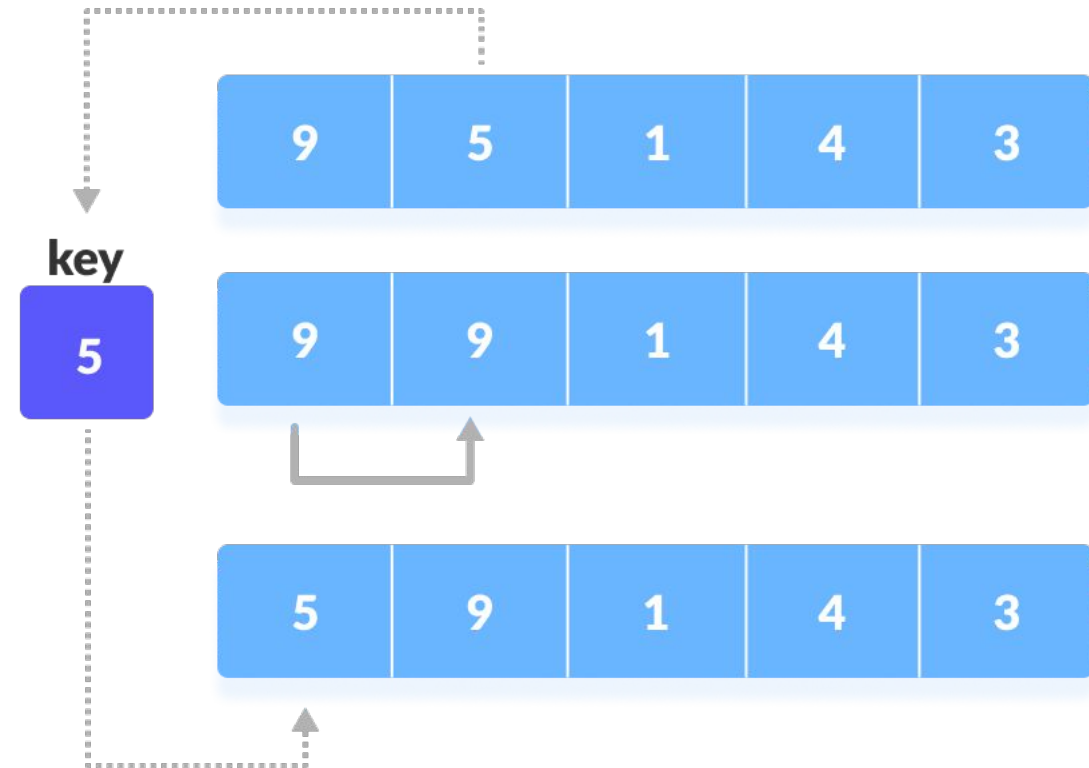
Compare with other elements

Insertion Sort

Insertion sort is a sorting algorithm that places an unsorted element at its suitable place in each iteration.

**Step 1 – If it is the first element, it is already sorted. return 1;
Step 2 – Pick next element
Step 3 – Compare with all elements in the sorted sub-list
Step 4 – Shift all the elements in the sorted sub-list that is greater than the value to be sorted
Step 5 – Insert the value
Step 6 – Repeat until list is sorted**

step = 1



```
def insertionSort(array):
```

```
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1
```

```
        # Compare key with each element
        on the left of it until an element smaller
        than it is found
```

```
        # For descending order, change
        key<array[j] to key>array[j].
```

```
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1
```

```
        # Place key at after the element just
        smaller than it.
```

```
        array[j + 1] = key
```

```
def selectionSort(array, size):
```

```
    for step in range(size):
        min_idx = step
```

```
        for i in range(step + 1, size):
```

```
            # to sort in descending order,
            change > to < in this line
            # select the minimum element
            in each loop
```

```
            if array[i] < array[min_idx]:
                min_idx = i
```

```
            # put min at the correct position
            (array[step], array[min_idx]) =
            (array[min_idx], array[step])
```

```
def bubbleSort(array):
```

```
    # loop to access each array element
    for i in range(len(array)):
```

```
        # loop to compare array elements
        for j in range(0, len(array) - i - 1):
```

```
            # compare two adjacent elements
            # change > to < to sort in
            descending order
```

```
            if array[j] > array[j + 1]:
```

```
                # swapping elements if elements
                # are not in the intended order
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp
```